Practice workbook · Input, process, output and other computational models

Naming the three stages of the IPO model or pulling an input, a process and an output out of a described program or problem; recognising sequential, parallel and multi-agent models; splitting a calculation across parallel processors.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-01

1. Question 2 · part b.ii · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b.ii only.

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2 Programs can process numerical data.

(a) Identify the method used to check input data matches set rules.

(1)

- ☒ **A** Abstraction
- ☒ **B** Encryption
- ☒ **C** Iteration
- ☒ **D** Validation

(b) **Figure 2** shows a section from an employee payslip produced by a computer program.

Pay Rate	Hours		Total Pay
£ 18.75	19		£ 356.25
	Tax	20%	£ 71.25
		Income	£ 285.00

Figure 2

(i) Complete the table to give **one** data type shown in **Figure 2** and give an example value.

(2)

Data type	
Example	

(ii) Complete the table to give an input, a process and an output used in the program to produce the payslip.

(3)

Input	
Process	
Output	

1. Question 2 · part b.ii · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b.ii only.

(c) Numerical values can be represented in denary, binary and hexadecimal.

A program converts denary values in the range 0 to 255 into hexadecimal.

(i) The program is tested using appropriate integer values.

Complete the table to show an example of normal, boundary and erroneous test data for this program.

(2)

	Test data
Normal	
Boundary	
Erroneous	

(ii) The program uses subprograms to do the conversion.

Open **Q02cii** in the code editor.

Amend the code to complete the program to:

- convert the user input to integer
- check the input is between 0 and 255
- call the **getHex** subprogram, with the user input as an argument.

Do not add any further functionality.

Save your code as **Q02ciiFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 2 = 12 marks)

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1. Question 2 · part b.ii · 3 marks · Working and answer

Working and answer

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2. Question 1 · part a · 3 marks

2019 JUN · Paper 01

Further practice · Attempt part a only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

(2)

1 .....

.....

2 .....

.....

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2. Question 1 · part a · 3 marks · Working and answer

Working and answer

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3. Question 3 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

3 The central processing unit (CPU) and other hardware components work together with the operating system to run programs and store data.

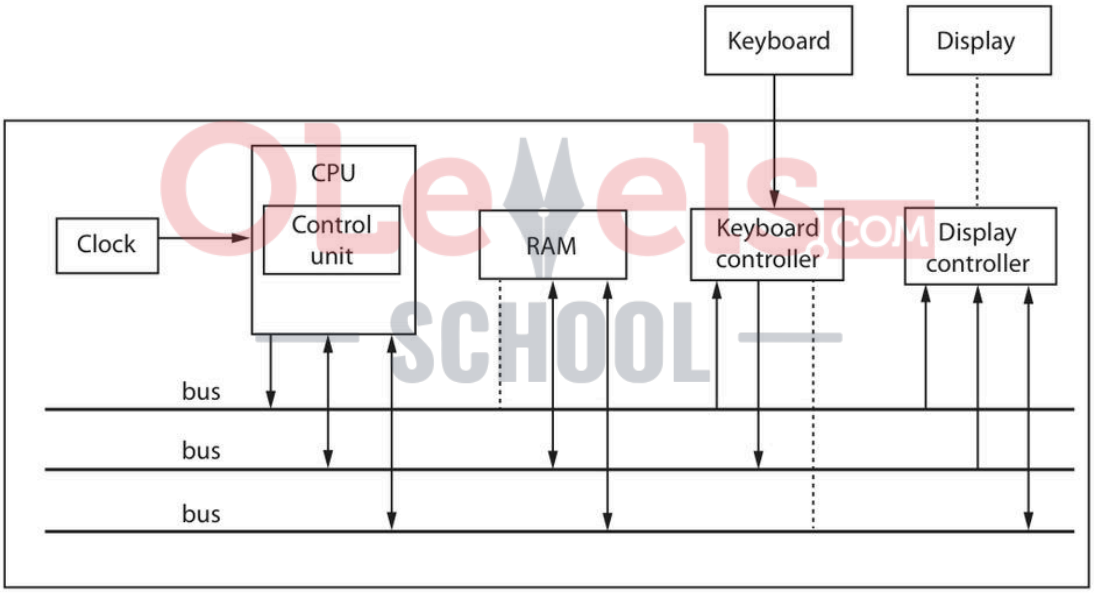
(a) Here is a partially completed diagram showing the hardware components used in the fetch-decode-execute cycle.

The function of the I/O controller is to manage communication between the CPU and input/output devices.

Complete the diagram by:

- labelling the buses
- adding directional arrows to the dotted lines to show the flow of communication.

(6)



(b) The CPU includes an area of cache memory.

Increasing the size of the cache will improve the performance of the CPU.

Explain how increasing the size of the cache improves the performance.

(2)

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3. Question 3 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

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(c) Some computers have more than one CPU. The CPUs work in parallel to execute instructions.

Complete the table to show how two CPUs working in parallel could execute the expression:

(5 * 2) + (4 * 3) – (24 / 4)

(4)

CPU 1	CPU 2
5 * 2 = 10	

(d) One function of an operating system is to manage files on secondary storage.

Describe how an operating system manages the storage of a file on random access secondary storage.

(4)

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(Total for Question 3 = 16 marks)

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3. Question 3 · part c · 4 marks · Working and answer

Working and answer

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4. Question 1 · part d · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part d only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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4. Question 1 · part d · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part d only.

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(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

.....

.....

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(ii) State the purpose of an assembler.

(1)

.....

(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

4. Question 1 · part d · 1 mark · Working and answer

Working and answer

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5. Question 5 · part a · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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5. Question 5 · part a · 2 marks

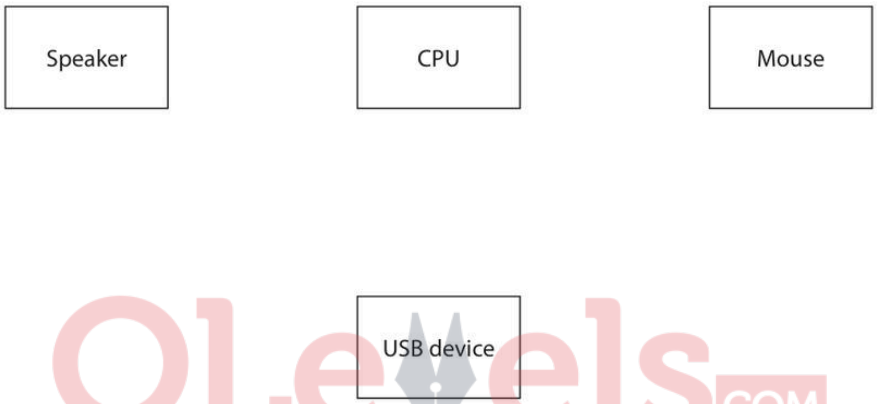
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Further practice · Attempt part a only.

(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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5. Question 5 · part a · 2 marks · Working and answer

Working and answer

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6. Question 1 · part a · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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6. Question 1 · part a · 3 marks

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Further practice · Attempt part a only.

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(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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6. Question 1 · part a · 3 marks · Working and answer

Working and answer

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7. Question 4 · part b · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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7. Question 4 · part b · 1 mark · Working and answer

Working and answer

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